

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -3

Aim

To study different types of network cables and prepare Ethernet straight-through and crossover cables using an RJ45 connector

Theory

Network cables connect devices and transmit data. The main types are:

Cable Type	Features	Application
UTP	Low cost, easy to install	Ethernet LAN
STP	Shielded, reduces interference	High-speed networks
Coaxial	Good shielding	TV and internet
Fibre Optic	High speed, long-distance transmission	High-speed networks

Tools Required

- Ethernet cable (CAT5/CAT5e)
- RJ45 connectors
- Crimping tool
- Wire stripper or cable cutter

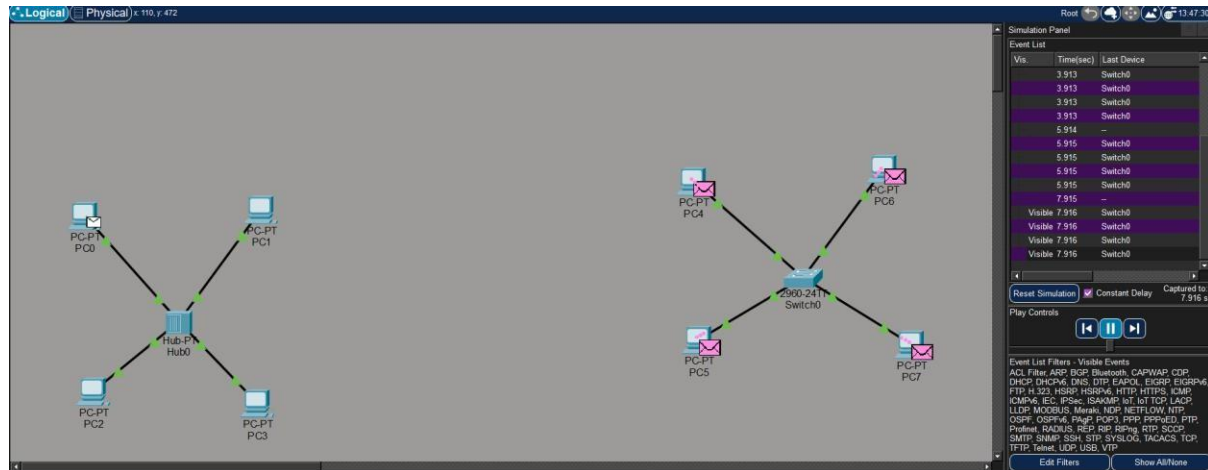
Difference Between Straight-Through and Crossover Cables

Straight-Through Cable	Crossover Cable
Same wiring standard at both ends	Different wiring standards at each end
Used to connect different types of devices	Traditionally used to connect similar devices directly
Example: PC to switch	Example: PC to PC

Part (c): Study of Different Types of Network Cables and Crimping

Procedure

1. Collect the Ethernet cable, RJ45 connectors, and crimping tool.
2. Cut the cable to the required length and strip approximately 1.5 cm of the outer covering.
3. Untwist and arrange the eight wires in the required colour sequence.
4. Straighten the wires and trim them evenly.
5. Insert the wires into the RJ45 connector in the correct order.
6. Use the crimping tool to secure the connector.
7. Repeat the process for the other end, using the same sequence for a straight-through cable or different sequences for a crossover cable.
8. Connect and test the cable.



Part (d): Analyse Network Devices Using Cisco Packet Tracer

Procedure

1. Open Cisco Packet Tracer.
2. Place four generic PCs and one Hub in the workspace.
3. Connect each PC to the Hub using copper straight-through cables.
4. Repeat the setup with four PCs connected to a Switch.
5. Assign IP addresses and subnet masks to the PCs through **Desktop** → **IP Configuration**.
6. Use the PDU tool to send a packet from one PC to another.
7. Observe packet forwarding in Simulation or Realtime mode.
8. Repeat the test for both the Hub and Switch networks.
9. Record the observations and compare how the Hub and Switch forward packets.

PC0

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 10.1.1.1

Subnet Mask: 255.0.0.0

Default Gateway:

DNS Server:

Result:

Thus the cisco packet tracer app is explored and simulations are done for hub and switch